

Lightweight AdaFace: Efficient Face Recognition Combining Quality-Aware Margin and Adaptive Knowledge Distillation

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Abstract—Mobile and edge face recognition demands efficient models, but lightweight networks often struggle with low-quality inputs. Traditional Knowledge Distillation (KD) can exacerbate this by forcing imitation of noisy samples. We propose Lightweight AdaFace, a novel framework combining adaptive class center updates with quality-aware margins. We utilize teacher features to dynamically correct class centers and adjust training margins based on student feature norms. Crucially, our Adaptive Geometry-aware mechanism penalizes student deviations only when teacher confidence is high, preventing overfitting to noise. Trained on MS1MV2, our method achieves competitive performance compared with existing lightweight models on LFW, IJB-C, and TinyFace, demonstrating superior robustness in low-quality recognition scenarios.

Keywords—Face Recognition, Knowledge Distillation, Lightweight Models, Edge Computing, Adaptive Margin

I. INTRODUCTION

In the era of maturing face recognition technology, deploying efficient and accurate models on resource-constrained devices (such as mobile phones and embedded systems) has become a critical challenge. While large models (e.g., IR-101) excel in various benchmarks, their massive parameter count and computational requirements make them unsuitable for edge applications. Conversely, lightweight models (e.g., MobileFaceNet [4]) offer speed but often suffer from significant drops in recognition rates when processing low-quality images that are low-resolution, blurred, or have large pose variations.

We observe two main limitations in existing solutions:

1. **Neglecting Image Quality:** Traditional loss functions (e.g., ArcFace [3]) or knowledge distillation methods typically treat all samples equally. However, forcing a lightweight model to learn "unrecognizable images" that even humans cannot identify often leads the model to memorize noise, thereby damaging its generalization ability for normal samples.

2. **Capacity Gap Between Teacher and Student:**

Directly using MSE Loss to force the student model to mimic the teacher model's feature space is overly rigid. Due to differences in model architecture and capacity, it is difficult for the student to fully replicate the teacher's feature distribution. This forced learning often yields suboptimal results.

Based on these observations, we propose a new training framework that allows a lightweight student model, under the guidance of a teacher model, to learn high-quality class features while dynamically adjusting learning difficulty based on image quality. Through an adaptive mechanism, the distillation intensity is automatically regulated to avoid overfitting to noise.

II. RELATED WORK

A. Margin-based Loss Functions

Deep face recognition has evolved significantly with the introduction of margin-based loss functions. SphereFace introduced multiplicative angular margins, while CosFace and ArcFace [3] proposed additive cosine and angular margins, respectively, to enhance intra-class compactness and inter-class separability. However, these methods typically assign a fixed margin to all samples, ignoring the variation in image quality. To address this, AdaFace [1] introduced an adaptive margin based on image quality, approximated by feature norms. It assigns larger margins to high-quality images and smaller margins to low-quality ones, preventing the model from overfitting to hard, unrecognizable samples. Our work adopts this quality-aware philosophy but integrates it into a lightweight distillation framework

B. Knowledge Distillation for Face Recognition

Knowledge Distillation (KD) is widely used to compress heavy models into lightweight ones. In face recognition, traditional KD methods often minimize the L_2 distance between teacher and student features. However, due to the capacity gap, strict feature mimicking can be detrimental. Recent works like AdaDistill [2] suggest relaxing this constraint by using teacher features to guide the student's class centers rather than forcing instance-level alignment. While AdaDistill effectively handles the capacity gap, it does not explicitly account for the geometric relationship between

teacher confidence and student alignment in the margin computation. Our method bridges this gap by introducing an adaptive geometry-aware penalty.

III. METHODOLOGY

We propose the Lightweight AdaFace framework, which combines the dynamic class center update strategy of AdaDistill [2] with the quality-aware margin mechanism of AdaFace [1]. We design a loss function named AdaptiveAdaFace, integrating the following techniques.

A. Adaptive Class Center Update

In traditional face recognition training, class centers (w_y) are randomly initialized and updated via backpropagation. To fully leverage the teacher model's knowledge, we employ an Exponential Moving Average (EMA) mechanism, using features extracted by the teacher model (f_t) to guide the update of class centers (w_{y_i}). The update formula is given by:

$$w_{y_i}^{(k)} = \alpha' \cdot w_{y_i}^{(k-1)} + (1 - \alpha') \cdot f_t^{(k)}, \quad (1)$$

where $w_{y_i}^{(k)}$ denotes the weight of the class center for class y_i at the k -th iteration, $f_t^{(k)}$ is the feature vector extracted by the teacher model, and α' is the dynamic weight coefficient balancing the previous center and the new teacher feature.

The coefficient α' is determined by two factors:

1. **Student-Teacher Similarity (α):** $\alpha = \cos(f_s, f_t)$. This reflects the student's understanding of the current sample. Low similarity implies the student has not learned well, resulting in a smaller α , forcing the class center closer to the teacher's feature (one-to-one imitation). As similarity increases, the center stabilizes towards the class average.
2. **Teacher Confidence (λ):** $\lambda = \cos(w_{old}, f_t)$. This represents the teacher's confidence in the current sample. If the sample is an outlier, λ is small, reducing the sample's impact on center updates.

The final coefficient is $\alpha' = \alpha \cdot \lambda$. This ensures that old centers are retained only when the "student understands" and the "teacher is confident"; otherwise, the teacher's features are used for correction.

B. Quality and Adaptive Geometry-aware Margin

Instead of a fixed margin, we generate a dynamic margin_scaler to adjust learning difficulty based on sample attributes.

- **Quality-aware Term (M_q):**

Following AdaFace theory, feature norm correlates with image quality. We use the normalized student feature norm (S_{norm}) as a proxy for quality to calculate the quality margin M_q :

$$M_q = -S_{norm} \cdot m, \quad (2)$$

High-quality images (high norm) result in a larger M_q , increasing the margin to force the learning of discriminative features. Low-quality images result in a smaller M_q , reducing the margin to prevent overfitting to noise.

- **Adaptive Geometry-aware Term (M_g)**

Early experiments with a static threshold for penalizing student-teacher discrepancies proved unstable and sensitive to hyperparameters. We improved this by introducing Teacher Confidence as a dynamic reference. We define two signals:

- Teacher Confidence ($Conf_t$): $\cos(w_y, f_t)$.
- Student-Teacher Alignment ($\cos_{st}$): $\cos(f_s, f_t)$.

The adaptive penalty is defined as:

$$Penalty = \text{ReLU}(Conf_t - \cos_{st}) \cdot Conf_t, \quad (3)$$

The penalty is triggered only when the student-teacher similarity is lower than the teacher's confidence. This means if the teacher is unsure (low $Conf_t$), we do not punish the student for deviating. The penalty strength is weighted by the teacher's confidence. The final geometry margin is $M_g = Penalty \cdot w$. This design eliminates manual tuning costs and improves robustness.

C. Final Adaptive Loss

The final logit calculation combines the above terms. We calculate the modified angle θ_m :

$$\theta_m = \theta + g_{angular}, \quad (4)$$

where the margin scaler combines quality and geometry terms. The final loss is formulated using the standard logits with these dynamic margins applied, ensuring the model is guided by image quality and dynamic geometric constraints from the teacher.

IV. EXPERIMENTS AND RESULTS

We used MS1MV2 (5.8M images) as the training set, IR-50 as the teacher model, and MobileFaceNet as the student model.

A. Training Dynamics & Stability Analysis

We analyzed the stability of the Adaptive Geometry-aware mechanism.

- **No Geometry-aware :** Without the teacher-driven geometry-aware constraint, the margin is determined solely by the student-side AdaFace mechanism. As a result, although image-quality-aware adaptation is preserved, no explicit student-teacher geometric alignment is enforced. Consequently, the logit-related metric fluctuates around zero without a clear convergence trend, exhibiting noise-like behavior during training.

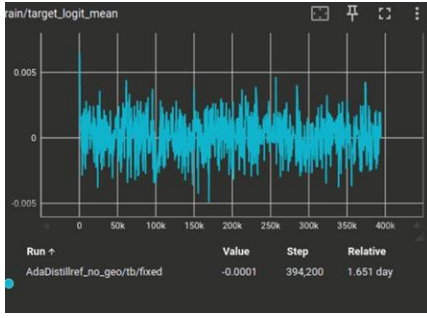


Fig. 1. Logic mean of static geometry-aware.

Static Geometry-aware: Early experiments exhibit a clear late-stage collapse, characterized by high-frequency oscillations after approximately 800k training steps. While static thresholds provide stable supervision during early training when student-teacher discrepancies are large, they fail to adapt during the convergence phase. As the student model performs fine-grained feature adjustments, a rigid fixed threshold frequently toggles the geometric penalty on and off, leading to numerical instability and oscillatory behavior.

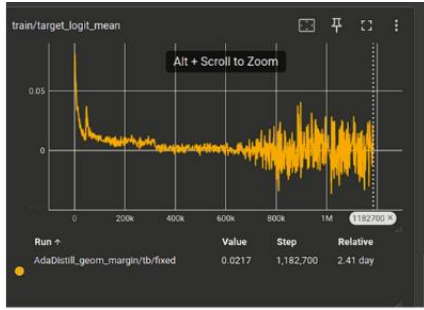


Fig. 2. Logic mean of static geometry-aware.

- **Adaptive Geometry-aware:** In contrast, the proposed adaptive geometry-aware mechanism shows a smooth and stable convergence trend. By using teacher confidence as a dynamic reference to modulate both the activation and strength of geometric penalties, oscillations are effectively eliminated, resulting in robust and stable geometric constraints throughout training.

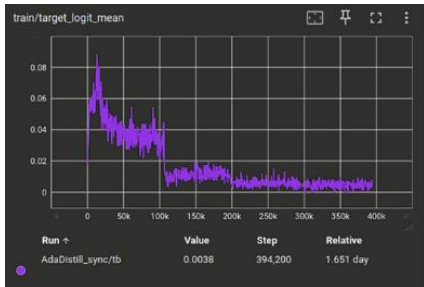


Fig. 3. Logic mean of adaptive geometry-aware.

B. Benchmark Results

We evaluated the method on LFW, AgeDB-30, CPLFW, IJB-C and TinyFace.

TABLE I. PERFORMANCE COMPARISON ON HIGH QUALITY DATASET

Method	Dataset			
	LFW	CFP-FP	AgeDB-30	CP-LFW
Baseline (MFN + ArcFace)	99.52	91.66	95.82	87.93
AdaDistill + ArcFace	99.47	95.14	96.68	89.50
AdaDistill + AdaFace	99.50	95.24	95.82	89.73
AdaDistill + AdaFace (Adaptive Geometry)	99.50	95.10	96.20	90.22

TABLE II. PERFORMANCE COMPARISON ON LOW QUALITY DATASET

Method	Dataset			
	IJB-C (1e-4)	IJB-C (1e-5)	TinyFace (Rank-1)	TinyFace (Rank-5)
Baseline (MFN + ArcFace)	89.13	81.65	--	--
AdaDistill + ArcFace	93.26	89.31	57.21	63.12
AdaDistill + AdaFace	93.82	90.00	58.69	64.03
AdaDistill + AdaFace (Adaptive Geometry)	93.98	90.33	59.31	64.43

On high-quality benchmarks such as LFW, CFP-FP, AgeDB-30, and CP-LFW, the proposed Adaptive Geometry-aware method maintains competitive performance without degradation, and achieves consistent improvements on more challenging cross-pose datasets such as CP-LFW.

On low-quality benchmarks, significant gains are observed on IJB-C at both FPR levels, demonstrating improved robustness under strict verification protocols. In particular, on the challenging low-resolution TinyFace dataset, our adaptive method achieves a Rank-1 accuracy of 59.31%, improving over the baseline by more than 2%. These results confirm that the proposed adaptive geometry-aware mechanism is especially effective in enhancing tail performance under noisy and low-quality conditions.

C. Teacher Model Comparison (IR-101 vs. IR-50)

Counter-intuitively, the smaller teacher (IR-50) produced a better student than the larger teacher (IR-101) (TinyFace: 58.69% vs. 57.47%). Analysis of training dynamics showed that IR-101's feature space was too complex for the lightweight student to mimic effectively. IR-50 provided "matched" capacity, allowing for more effective convergence and "teaching according to aptitude."

TABLE III. PERFORMANCE COMPARISON ON DIFFERENT TEACHER MODELS

Method	Dataset			
	LFW	CFP-FP	AgeDB-30	CP-LFW
IR101(teacher)	99.08	98.61	98.05	87.93
MFN (Student)	99.64	94.81	96.12	89.73
IR-50 (Teacher)	99.85	98.09	97.85	92.85
MFN (Student)	99.50	95.24	95.82	89.73

TABLE IV. PERFORMANCE COMPARISON ON DIFFERENT TEACHER MODELS

Method	Dataset		
	<i>IJB-C (1e-4)</i>	<i>IJB-C (1e-5)</i>	<i>TinyFace (Rank-1)</i>
Ir101(teacher)	96.81	94.58	68.11
MFN (Student)	92.21	88.32	57.47
IR-50 (Teacher)	96.20	93.06	64.99
MFN (Student)	93.82	90.00	58.69

V. REAL-WORLD IMPLEMENTATION

To verify feasibility, we deployed the trained model on a Raspberry Pi for a "Face Recognition Check-in System." The model was converted to ONNX format and accelerated using ONNX Runtime. Results showed that the model maintained high accuracy while significantly reducing computational load, enabling real-time face recognition and tracking on edge devices.

VI. CONCLUSION

In this work, we present the Lightweight AdaFace, a multi-level adaptive distillation framework for lightweight face recognition. The key insight of this work is that stable and effective distillation under low-quality and challenging conditions requires joint adaptive regulation of both class centers and classification margins, rather than adapting either component in isolation.

At the center level, we introduce an adaptive center update mechanism that leverages teacher confidence and student-teacher similarity to selectively guide class center evolution, preventing unstable center shifts during early training stages and on difficult samples. At the margin level, we combine two complementary adaptive strategies: quality-aware margin modulation from AdaFace to account for image quality variations, and a confidence-driven adaptive geometry mechanism that dynamically controls the activation and strength of geometric penalties. Together, these mechanisms ensure that supervision is applied only when it is reliable and informative.

Extensive experiments on both high-quality and low-quality benchmarks demonstrate that the proposed framework consistently improves training stability and recognition performance, particularly under noisy and low-resolution scenarios. Moreover, our results highlight that effective distillation depends on coordinated center-margin adaptation and appropriate teacher selection, rather than relying solely on stronger teachers. Without increasing inference-time complexity, Lightweight AdaFace provides a robust and practical training paradigm for deploying lightweight face recognition models in resource-constrained and edge environments.

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